1

loop: flw f1, 0(x1) → IF ID Ex M WB

fmul f9, f1, f0 → IF ~~ID~~p ID M1 M2 M3 M WB LDC

fmuls f3, f2, f3 → EST ~~IF~~p IF ID M1 M2 M3 M WB

fmuls f0, f1, f4 → IF ~~ID~~p ID M1 M2 M3 M WB

fadds f0, f2, f4 → EST ~~IF~~p IF ~~ID~~p ID A1 A2 M WB

add: x2, x1, x4 → EST ~~IF~~p IF ~~ID~~p ID Ex M WB

bne x2, x2, loop → ~~IF~~p IF ID Ex M WB

add: x1, x1, x4 → IF ID ~~Ex~~ ~~M~~ ~~WB~~

add: x2, x2, 8 → IF ~~ID~~ ~~Ex~~ ~~M~~ ~~WB~~

flw → IF ID Ex M WB

fmuls → IF ~~ID~~p ID M1 M2 M3 M WB

b)

$N^o\ CICLOS = 30 \cdot 13 + 4 = 394$

$N^o\ INSTR = 30 \cdot 7 + 2 = 212$

$CPI = \frac{N^o\ CICLOS}{N^o\ INSTR} = \frac{394}{212} = 1{,}86$

2

M → add x3, x4, x7

EX → addi x3, x3, 1

(M)

RegWrite M = 1

ResultSrc M = 00

MemWrite = 0

(Ex)

RegWrite E = 1

ResultSrc = 00

MemWrite = 0

Jump E = 0

Branch E = 0

ALUControlE = 000

ALUSrc E = 1

Forward AE = 10

Forward BE = 00 ó XX

b) if(RS1E == RdM) && RegWriteM && (Rs1E != 0) → Forward = 10

else if (RS1E == RdW) && → APUNTES

3

$MP = 1\ Mb = 2^{20}$

$MC = 8\ Kb = 2^{13}$ } 16 bl

$LÍNEA = 512\ b = 2^9$

a) DF | ETIQ 7 | BL 4 | DESP 9 |

DF | ETIQ 7 | BL 4 | DESP 9 |

0x23345 → 0010 0011 0011 0100 0101

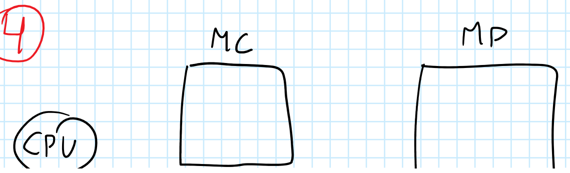
0x713BF → 0111 0001 0011 1011 1111

0x71584 → 0111 0001 0101 1000 0100

Nº BL	0x23345	0x713BF	0x71584
1001	001 0001(0x11) F	011 1000 (0x38) F	011 1000 (0x38)
1010	001 0001(0x11) P	011 1000 (0x38) P	011 1000 (0x38) A

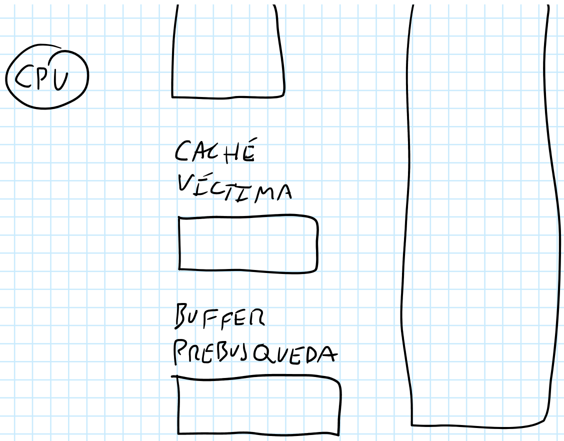
CJTO	BL	0x23345	0x713BF	0x71584
001	0	0010 0011 (0x23) F	0010 0011 (0x23)	—
001	1		0111 0001 (0x71) F	—
010	0	0010 0011 (0x23) P	0010 0011 (0x23)	—
010	1		0111 0001 (0x71) P	— A

4



1. Acierto en caché

2a. Fallo en caché? Intercambio de bloques



1. Acierto en caché

2a. Fallo en caché, Acierto en CV } Intercambio de bloques $C \to CV$ / $CV \to C$

2b. Fallo en caché, Fallo en CV, Acierto en buffer } $B \to C$, $C \to CV$

2c. Fallo en caché, Fallo en CV, Fallo en buffer ⇓ Busqueda en MP } (BLN) $MP \to C$, $C \to CV$, $MP(BLN) \Rightarrow BUFFER$

Hoja tema 4

① 200 MIPS
$T_{ESPERA} = 5\,ms$

$$200 \cdot 10^6 \ \frac{inst}{sec} \cdot 5 \cdot 10^{-3} = 10^6 \text{ instrucciones}$$

② CICLOS PARA LECTURA = $30 \cdot 2000 = 60000$ ciclos/sec

% CICLOS DE CPU CONSUMIDOS EN LA LECTURA = $\frac{60000}{2{,}7 \cdot 10^9} = 0{,}002$ %

③ 1 GHz
$CPI = 8$

a) RÁFAGA

INST	DMA	DMA	...	DMA	INST+1

$$Velocidad = \frac{10^9 \ ciclos/sec}{1 \ ciclo/pal} = 10^9 \ pal/sec$$

b)

INST	DMA	INST+1	DMA

$$Velocidad = \frac{10^9 \ ciclos/sec}{9 \ ciclos/pal} = 1{,}11 \cdot 10^8 \ pal/sec$$

c) TRANSPARENTE

$$Velocidad = \frac{10 \ ciclos/sec}{4 \ ciclos/pal} = 0{,}25 \cdot 10^9 \ pal/sec$$

8

INST	INST+1

DMA DMA DMA DMA

④ PERIFÉRICO VEZ. TRANSF = $2 \cdot 10^6$ bytes
$T_{inst} = 100$ ms
Inicialización DMA = 10 inst
Transf de palabra = 50 ns
Transferimos = 512 bytes

$$T_{BUS} = \frac{512 \ bytes}{4 \ bytes/pal} \cdot 50 \ ns/pal = 6400 \ ns$$

$$T_{trans\_byte} = \frac{1}{2 \cdot 10^6} = 0{,}5 \cdot 10^{-6} \ sec$$

$$T_{trans\_pal} = 4 \cdot 0{,}5 \cdot 10^{-6} = 2 \cdot 10^{-6} \ sec$$

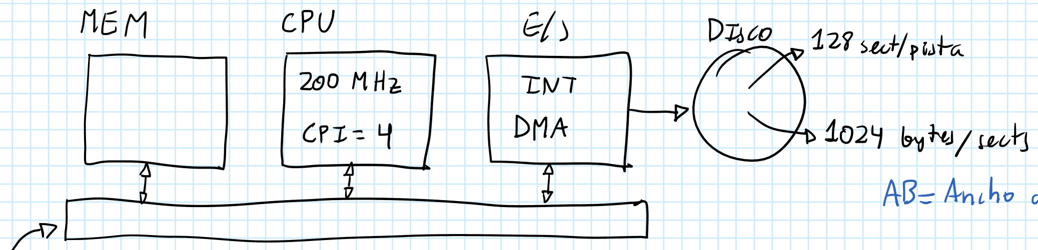
$$T_{periférico} = 512 \ bytes \cdot 0{,}5 \cdot 10^{-6} \ sec = 256 \cdot 10^{-6} \ sec$$

$$T_{CPU\_PARA\ HACER\ OTRAS\ COSAS} = 256 \cdot 10^{-6} - 6{,}4 \cdot 10^{-6} = 249{,}6 \cdot 10^{-6} \ sec$$

$$N^{o} \ INST = \frac{249{,}6 \cdot 10^{-6}}{100 \cdot 10^{-9}} = 2496 \ INST$$

⑤ MEM CPU E/S DISCO

5



AB = Ancho de Banda

$T_{rec\_int} = 100\ ns$

8 bytes/inst

$RUT\_INT = 20$ inst

a) $AB_{SIST\_INT} = \frac{n^o\ bytes}{T_{INTERRUP}} = \frac{8\ bytes}{T_{REC} + T_{RTC}} =$

$= \frac{8\ bytes}{100\ ns + (20\ inst \cdot 4\ ciclos/inst) + 5\ ns/ciclo} = \frac{8}{500} = 16\ Mb/s$

b)

$AB_{SIST\_INT} >= AB_{DISCO}$

$AB_{DISCO} = \frac{n^o\ bytes}{pista} \cdot \frac{n^o\ pistas}{sec} = 2^7 \cdot 2^{10} \cdot w$

$16 \cdot 10^6\ bytes/sec \geq 2^{17} w \Rightarrow w \leq \frac{16 \cdot 10^6}{2^{17}} = 127{,}07\ rps$

$rpm = 127{,}07 \cdot 60 = \underline{7324{,}21}$

c)

INST	DMA	INST+1	DMA
4	1	4	1

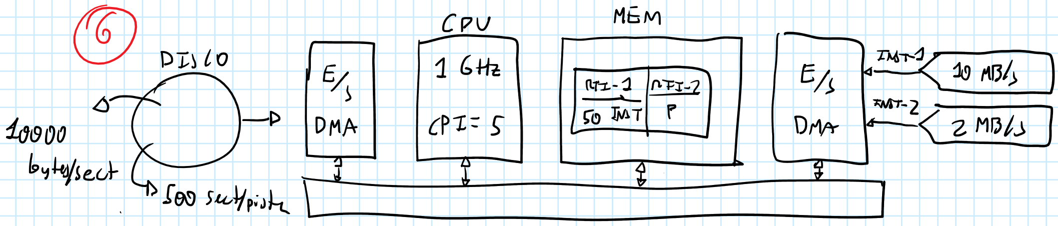
$T_{ROBADO} = N^o T_{ransf} \cdot T_{transf} = \frac{1024\ bytes/sect}{8\ bytes/trans} \cdot 5\ ns = \underline{640\ ns}$

d) $T_{LECTURA} = \frac{1}{127{,}07} = 8157 \cdot 10^{-6}\ s$

$T_{LECTURA\_SECTOR} = \frac{8157 \cdot 10^{-6} s}{128} = 64 \cdot 10^{-6}\ s/sector$

$\%\ TIEMPO\_CPU\_DEDICA\ A\ E/S = \frac{640 \cdot 10^{-9}}{64000 \cdot 10^{-9}} = \underline{1\ \%}$

6



a) $1\ GHz = 1\ ns$

$T_{INS} = 5 \cdot 1\ ns = 5\ ns$

$N^o\ INT1 = \frac{10 \cdot 10^{-6}\ bytes}{4\ bytes/int} = 2{,}5 \cdot 10^6\ int/sect$

$T_{INT} = 2{,}5 \cdot 10^6\ int/sect \cdot 50\ inst/int \cdot 5\ ns = 0{,}625\ sec$ (interrupciones)

$0{,}3751$ PARA INT 2

$N^o\ INT2 = \frac{2 \cdot 10^6\ bytes/sec}{4\ bytes/int} = 0{,}5 \cdot 10^6\ int/sec$

$0{,}3751 = N^o_{INSTR}/INT \cdot 0{,}5 \cdot 10^6 \cdot 5\ ns \Rightarrow \boxed{N^o INST = \frac{0{,}3751}{2{,}5 \cdot 10^{-5}} = \underline{150}}$

b) $CPI = 5 + 1\ ROBO = 6\ CICLOS$

$T_{DMA} = \frac{1}{6} = 0{,}17\ sec$

$T_{PARA\ INT2} = 1 - 0{,}3751 - 0{,}17 = 0{,}205 \Rightarrow N^o INST = \frac{0{,}205}{2{,}5 \cdot 10^{-5}} = \underline{82\ inst}$

c) ROBO DE CICLO

$$AB_{DMA} = \frac{4 \text{ bytes}}{(5+1)\,\text{ciclo} \cdot 10^{-9}\, \frac{\text{sec}}{\text{ciclo}}} = 666{,}6 \text{ MB/sec}$$

$$AB_{DISCO} = \frac{N^o \text{ bytes}}{\text{pista}} \cdot \frac{N^o \text{ pista}}{\text{sec}} = 5 \cdot 10^6 \cdot \frac{W}{60}$$

$$AB_{DMA} \geq AB_{DISCO} \Rightarrow 666{,}6 \cdot 10^6 \geq 5 \cdot 10^6 \frac{W}{60} \Rightarrow \underline{W \leq 8000 \text{ rpm}}$$

RÁFAGA

$$AB_{DMA} = \frac{4 \text{ bytes}}{1 \text{ ciclo} \cdot 10^{-9} s/\text{sector}} = 4 \text{ GB/sec}$$

$$AB_{DMA} \geq AB_{DISCO} \Rightarrow 4 \cdot 10^9 \geq 5 \cdot 10^6 \frac{W}{60} \Rightarrow \underline{W = 48000 \text{ rpm}}$$

Teoría

Redondeo

$+ \boxed{1{,}011110}\,00$ → 1,01110

01 → 1,01110

10 → 1,01110

11 → 1,011111

Punto flotante:

(1)

Expresar en PF:

i) $+0 \times 8452 \Rightarrow +1000\ 0100\ 0101\ 0010 \Rightarrow 1{,}000\ 0100\ 0101\ 0010 \cdot 2^{15}$

ii) $133 \Rightarrow 10000101 \cdot 2^0 \Rightarrow 1{,}0000101 \cdot 2^7$

Expresar en decimal

i) 0x BB50_0000

ii) $0\times 8040\_0000 \Rightarrow 1|000\ 0000\ 0|100\ 0000\ 0000\ 0000\ 0000$

SIGNO = −
EXP = 126
MANTISA = 0,1

$\} \ 0{,}1 \cdot 2^{-126}$

$[1|000\ 0000\ 0|111 \ldots 111]$

$\hookrightarrow 0{,}\underbrace{11\ldots11}_{23} \cdot 2^{-126} = (1-2^{-23}) \cdot 2^{-126}$

iii) $0\times 80C0\_0000 \Rightarrow 1|000\ 0000\ 1|100\ 0000\ 0000\ 0000\ 0000$

SIGNO = −
EXP = 1-127 = -126
MANTISA = $(1{,}1)_2 = (1{,}5)_{10}$

$\} \ -1{,}5 \cdot 2^{-126}$

iv) $0\times FF80\_0000 \Rightarrow 1|111\ 1111\ 1|000\ 0000 \ldots 0 = -\infty$

v) $0\times 40F0\_0000 \Rightarrow 0|100\ 0000\ 1|111\ 0000 \ldots 0$

SIGNO = +
EXP = 129 − 127 = 2
MANT = 1,111 = 1,875

$\} \ 1{,}875 \cdot 2^2$

Hoja 4

(9)

NPV DESP

TLB

TABLA DE PÁGINAS

Hoja 4

9

$MP = 4KB \rightarrow 2^{12}$ → DV: NPV 3 | DESP 10

$T_{PÁG} = 1KB \rightarrow 2^{10}$ → DF: NPF 2 | DESP 10

$MV = 8KB \rightarrow 2^{13}$

REF → 0x144C → 0x1F04 → 0x0750 → 0x008C

TLB

NPV	NPF	LRU
0	3	0
1	1	1

TABLA DE PÁGINAS

NPV	VAL	LRU	NPF
0	1	0	3
1	1	1	1
2	0	–	0
3	0	–	
4	0	–	1
5	1	3	2 2(0)
6	0	–	2
7	1	2	0 (2)

realmente no sé cual debería ser

a)
0x144C → [1 01|00] 0100 1100
0x1F04 → [1 11|1] 1 0000 0100
0x0750 → [0 01|1] 1 0101 0000
0x008C → [0 00|0] 0 1000 1100

b)
0x0F6C → 0 1111 0110 1100
0x120C → 1 0010 0000 1100
0x0478 → 0 0100 0111 1000

0x0F6C → FALLO PÁG

NPV	VAL	LRU	NPF
0	1	1	3
1	1	2	1
2	0		
3	1	0	2
4	0		
5	0	3	2
6	0		
7	1	3	0

TLB

NPV	NPF	LRU
0	3	1
3	2	0

⇒

0x120C → FALLO PÁG

NPV	VAL	LRU	NPF
0	1	1 2	3
1	1	2 3	1
2	0		
3	1	0 1	2
4	0 1	0	0
5	0	3	2
6	0		
7	1 0	3	0

TLB

NPV	NPF	LRU
0 4	3 0	1 0
3	2	0 1

⇒

0x0478 → ACIERTO

NPV	VAL	LRU	NPF
0	1	3 1 2	3
1	1	0 2 3	1
2	0		
3	1	2 0 1	2
4	0 1	1 0	0
5	0	3	2
6	0		
7	1 0	3	0

TLB

NPV	NPF	LRU
0 4	3 0	1 0 1
3 1	2 1	0 1 0

c) CACHE → 8 bl · 256 bytes = 2KB = 2^{11} → No puede ser mayor que el tamaño de página (2^{10})

DF: [2 | 8] (11) → DF: [2 | 8] (10) → No se puede usar un emplazamiento directo → DF: ETIQ 2 | CJTO 2 | DESP 8

d) 0x1C00 - 0x1FFF

NPV	NPF
7	3
4	1

0x1C00 → 111 0000000000

0x1FFF → 111 1111111111

DF → 11 00 00000000
11 11 11111111

REF ?

$MC = 1KB$

$BL = 256 KB$

DIRECTA

NPF 2 | DESP 10

ETIQ 2 | CJTO 2 | DESP 8

DF

MARCO	ETIQ
0	11
1	01
2	11
3	11

01 01 xx...x → DF = 0x5xx → 0x500, 0x5FF

DV = 100 01 xxxx xxxx → 0x1100, 0x11FF